

Successor-Saturated Ordinals and Indefinite Tail Activation

A RED/DED Operator Extension of the Successor-Closed Sequence Lexicon

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Abstract

This paper strengthens a lexical hypothesis about ordinal meaning: encoding a position as FIRST necessarily encodes a successor position, and more strongly, encodes all successor positions in the relevant sequence. Formally, sequence is defined as an index structure whose domain is successor-closed (or at minimum two-element) and whose semantic interface guarantees a successor for the initial index. Under the Successor-Closed Sequence Lexicon (SCSL), FIRST $\Rightarrow$ SECOND is not pragmatic enrichment but a lexical entailment/presupposition triggered by the ordinal itself. Building on SCSL, I introduce Indefinite Tail Activation (ITA): when FIRST anchors the beginning of a sequence, all remaining successor slots are introduced into the discourse as an underspecified remainder that is canonically lexicalized as non-specific indefiniteness (“some / a few / others”). ITA is not derived from conversational reasoning; it is a lexical interface consequence of successor saturation. The proposal is integrated into a RED/DED operator architecture: RED implements refinement along the successor chain (FIRST $\rightarrow$ SECOND $\rightarrow$ THIRD $\rightarrow$...), while DED implements controlled collapse from refined ordinal microstructure back to a coarse remainder expressed by non-specific indefinites. The resulting system intentionally changes the diagnostic landscape: standard implicature tests (cancellability, weakening under embedding) are not preserved, because successor availability and tail-indefiniteness are lexical definedness/interface conditions rather than conversational inferences. The paper concludes with a replication-ready diagnostic set, a compact elicitation grid, and computational/neuroscientific implications for successor-chain activation and ordinal-driven discourse updates.

Keywords: ordinals; successor closure; lexical entailment; presupposition/definedness; indefiniteness; some; remainder; RED diagnostic; DED collapse; refinement/coarsening; discourse update; ellipsis/anaphora; generalized quantifiers; crosslinguistic elicitation (Barwise & Cooper 1981; Keenan & Stavi 1986).

Novelty & Contributions (Reviewer-Facing Summary)

Contribution 1 (Lexical semantics). FIRST is treated as a successor-saturating trigger: if FIRST is interpretable, the successor function at the initial index is defined ($\text{FIRST} \Rightarrow \text{SECOND}$), and the successor chain is available as a structured tail of indices.

Contribution 2 (New interface effect). Indefinite Tail Activation (ITA): mentioning FIRST introduces a remainder discourse resource corresponding to the tail, canonically lexicalized as non-specific indefiniteness (“some / a few / others”), independent of Gricean enrichment.

Contribution 3 (Bidirectional operators). The paper composes successor refinement (RED) with controlled collapse (DED) to derive a principled refinement/coarsening path: $\text{FIRST} \rightarrow \text{SECOND} \rightarrow \dots \rightarrow \text{TAIL} \rightsquigarrow \text{SOME}(\text{others})$.

Contribution 4 (Diagnostic redesign). Because successor and tail-indefiniteness are lexical definedness/interface conditions, classic implicature diagnostics (cancellability, weakening under embedding) are predicted to fail in targeted ways; the theory replaces rather than preserves the standard diagnostic regime.

Contribution 5 (Measurable predictions). The proposal yields testable predictions about (i) successor-cancellation degradation, (ii) enhanced naturalness of “some others” continuations after FIRST (ITA), and (iii) reduced cost of refining tail references into specific successors (SECOND/THIRD/...).

Contribution 6 (Cross-domain link). The remainder variable introduced by FIRST bridges ordinal structure to non-specific indefinite quantification, enabling explicit tests at the interface of ordinal semantics, discourse update, and quantifier choice.

1. Introduction

A position in a sequence is not semantically solitary. If an expression encodes a token as FIRST, the structure that makes FIRST meaningful already encodes successor structure—at minimum a SECOND slot. The Successor-Closed Sequence Lexicon (SCSL) captures this by treating successor availability as a lexical definedness condition for FIRST, yielding $\text{FIRST} \Rightarrow \text{SECOND}$ as a lexical consequence rather than a pragmatic enrichment. This paper adopts SCSL and strengthens it: FIRST is taken to encode not only the next successor, but the entire successor

chain, and to introduce that chain into discourse as an underspecified remainder that is canonically lexicalized as non-specific indefiniteness (“some / a few / others”).

Methodologically, the proposal does not aim to preserve standard implicature diagnostics. If successor availability and tail-indefiniteness are lexical definedness/interface effects, cancellability and weakening under embedding are not expected to behave like implicatures. The theory is therefore evaluated by its operator-driven predictions and by the stability of its RED/DED diagnostics.

2. Related Work and Positioning

The proposal intersects three traditions: (i) ordinal semantics (including analyses relating ordinals to ordering structure and superlative-like mechanisms), (ii) generalized quantifier theory and indefinite semantics, and (iii) dynamic/discourse frameworks that model anaphora and continuation licensing. The present approach is not a general claim about discourse representation; rather, it lexicalizes successor structure and remainder availability inside FIRST itself, and composes this with a RED/DED operator architecture that treats refinement and collapse as explicit, measurable procedures. (Ionin & Matushansky 2006; Barwise & Cooper 1981; Keenan & Stavi 1986; Heim 1982; Kamp & Reyle 1993; Partee 1987; Yee 2010). (Panner 2026b). A generalized-quantifier-only baseline can model quantifier strength but does not, by itself, predict that encountering FIRST must introduce a successor chain or a remainder discourse object. A dynamic/DRT-style baseline can represent a sequence if independently introduced, but does not force the successor chain and remainder variable to be introduced as a lexical consequence of FIRST. The distinguishing content of the present account is therefore the successor-saturation + ITA package, and its diagnostic consequences. (Barwise & Cooper 1981; Keenan & Stavi 1986; Partee 1987).

3. SCSL+ and Successor Saturation

3.1 Sequence objects

Let a sequence be an index structure $\Sigma = \langle I, <, \text{succ}, \text{min} \rangle$. Under SCSL, FIRST is defined only if $\text{succ}(\text{min}(I))$ is defined; minimally, this requires $|I| \geq 2$. SCSL+ strengthens this to successor

saturation: the active domain is successor-closed (relative to the relevant segment), and every index in the active segment is reachable from $\min(I)$ by iterated successor.

$\Sigma = \langle I, <, \text{succ}, \min \rangle$, with SCSL definedness: $\text{succ}(\min(I)) \downarrow$ (Panner 2026a).

Successor tail: $\text{TAIL}(\Sigma) = \{ \text{succ}^k(\min(I)) \mid k \geq 1 \text{ and } \text{succ}^k(\min(I)) \downarrow \}$

3.2 Indices vs occupants

To avoid trivializing existential commitments about individuals, distinguish the index spine from its occupancy. Let $f : I \rightarrow D$ be a partial mapping from indices to individuals/events. The successor-saturation claims are constraints over the index spine I , not over the occupancy of each index. This separation allows “empty slot” readings while preserving successor structure as a lexical requirement.

4. Indefinite Tail Activation (ITA)

4.1 Tail remainder as a discourse resource

Indefinite Tail Activation adds a mandatory interface effect: interpreting FIRST introduces a remainder variable t corresponding to $\text{TAIL}(\Sigma)$. This variable is introduced non-specifically: the discourse does not commit to identifying which successor indices will be referenced later. The canonical surface lexicalization of this remainder is a non-specific indefinite (“some / a few / others”).

ITA update: $\text{FIRST} \Rightarrow$ introduce t , with t ranging over $\text{TAIL}(\Sigma)$ non-specifically

Remainder operator: $\kappa \exists (t) \equiv \exists X (X \subseteq t \wedge |X| \geq 1 \wedge \text{Card}(X) \text{ underspecified})$

4.2 Why ITA is not implicature

On this account, the availability of a successor chain and of a remainder (“some others”) is not derived from conversational reasoning about why a speaker used first. It follows from lexical definedness and a lexical interface rule. Therefore, classic implicature diagnostics—especially cancellability—are predicted to fail in targeted ways.

4.3 Compositional derivation: FIRST licenses “some others” via ITA

This subsection provides an explicit compositional sketch for the flagship discourse “John was first. Some others followed.” The goal is not to fix a single formalism, but to make the SCSL+ + ITA mechanism checkable under standard compositional assumptions. (Heim 1982; Kamp & Reyle 1993; Barwise & Cooper 1981; Partee 1987). Assume a contextual sequence object $\Sigma = \langle I, <, \text{succ}, \text{min} \rangle$ with an (optionally partial) occupancy mapping $f: I \rightarrow D$, distinguishing index availability from occupant realization. Let FIRST denote an index selector with a definedness condition plus an interface update: Definedness: $\llbracket \text{first} \rrbracket (\Sigma)$ is defined only if $\text{succ}(\text{min}(I)) \downarrow$ (and, under successor saturation, a non-empty tail exists). Value: $\llbracket \text{first} \rrbracket (\Sigma) = \text{min}(I)$. Update (ITA): introduce a remainder variable t ranging over $\text{TAIL}(\Sigma) = \{\text{succ}^k(\text{min}(I)) \mid k \geq 1 \wedge \text{succ}^k(\text{min}(I)) \downarrow\}$, treated non-specifically. (Panner 2026a).

Sentence (i) “John was first” asserts that John is associated with the initial index (e.g., $\text{John} = f(\text{min}(I))$ where defined), while triggering ITA. Sentence (ii) interprets “some others” as $\kappa \exists (t)$, an existential remainder operator over the tail, yielding $\exists X (X \subseteq \text{range}(f|_t) \wedge |X| \geq 1 \wedge \text{Card}(X)$ underspecified), and predicates FOLLOW(X) (or a plural follow relation) of that remainder set. Because earliest lacks the lexical ITA update, the theory predicts a stable continuation advantage after FIRST relative to earliest in matched controls. (Kamp & Reyle 1993; Yee 2010).

4.4 Embedding and projection: modals and negation

SCSL and ITA are treated as lexical definedness/interface effects; accordingly, they are expected to project more like presuppositions than like cancellable implicatures. Two key embedding environments are modals and negation. (Beaver 2001; Heim 1982).

Modal embedding. In “John might be first,” the truth-conditional predication is evaluated in accessible worlds, but interpretability of the ordinal still requires a successor-bearing sequence object. Therefore, successor availability—and the remainder variable introduced by ITA—remain active under modalization (subject to local accommodation strategies). (Kratzer 1991; Beaver 2001).

Negation. In “John is not first,” negation targets the predication about John’s association with the initial index; it does not erase the definedness conditions contributed by FIRST. Thus, successor and tail commitments are predicted to persist unless a repair/coercion strategy is

invoked, which is precisely why implicature diagnostics (e.g., cancellability) are not preserved. (Beaver 2001; Kamp & Reyle 1993).

5. RED/DED: Refinement to successors and collapse to “some”

5.1 RED as successor refinement

RED implements refinement along the successor chain. Define $\text{Next}\Sigma(i) = \text{succ}(i)$ wherever $\text{succ}(i)$ is defined. Then SECOND is the successor-refinement of FIRST, and in general the $(k+1)$ th position is obtained by iterated successor from the initial anchor.

$$\text{Next}\Sigma(i) = \text{succ}(i); \text{ hence } \llbracket \text{second} \rrbracket (\Sigma) = \text{Next}\Sigma(\llbracket \text{first} \rrbracket (\Sigma))$$

5.2 DED as controlled collapse to the remainder

DED implements controlled collapse from refined ordinal microstructure to a coarse remainder. The collapse operation maps specific successor identities into the remainder variable t , yielding a non-specific indefinite summary (“some/a few/others”) without deleting the underlying successor spine.

6. Diagnostic consequences (non-preservation of standard pragmatics tests)

Because successor and tail-indefiniteness are lexical definedness/interface effects, classic implicature diagnostics are not preserved. In particular, attempts to cancel successor structure after asserting FIRST are predicted to yield contradiction or definedness failure rather than a successful implicature cancellation.

- (1) # John was first—indeed there was no second position.
- (2) # John was first—indeed there were no other positions.
- (3) John was first. Some others followed.
- (4) John was first. A few others followed.
- (5) John was first. Some others followed—specifically, the second was Mary.

7. Distinguishing predictions

The account predicts (i) strong degradation of successor- and tail-cancellation, (ii) heightened acceptability of “some others” continuations following FIRST relative to controls (ITA), and (iii) reduced cost for refining tail references to specific successors (SECOND/THIRD/...) relative to baselines that do not lexicalize the successor chain.

7.1 Baseline-failure demonstration (MP-family prediction contrasts)

Table 7A states discriminating predictions against two common baselines (GQ-only and DRT/dynamic). The aim is to make the contribution falsifiable without requiring immediate large-scale data collection: each MP-family is paired with (i) a baseline divergence point and (ii) an explicit falsifier that would force revision or rejection of the lexical ITA/SCSL+ claims.

Table 7A. Baseline-failure predictions for ITA minimal-pair families (discriminating observations).

7.2 Falsifiability propositions

Falsifier 1 (ITA): If FIRST does not increase continuation naturalness for “some/a few/others” relative to earliest controls across languages and items, ITA is unsupported.

Falsifier 2 (SCSL definedness): If cancellation (“first... but there was no second position / no others”) is not reliably degraded beyond controls, the successor-closure definedness condition is too strong and must be weakened or re-scoped.

MP family	SCSL+ITA prediction	GQ-only baseline prediction	DRT/dynamic baseline prediction	Predicted signature under SCSL+ITA	Where baseline fails	Discriminating observation (falsifier)
MP-ITA5–6 (cancellation)	Cancellation contradicts lexical successor/tail definedness → sharp SR degradation.	At most pragmatically odd; predicts weaker SR penalty and higher acceptability.	Repairable discourse structure; predicts weaker SR penalty.	SR ratings reliably lower than Ctrl; entailment coding for successor/tail remains high in SR.	Baselines do not force contradiction-like degradation tied specifically to FIRST.	If SR cancellation is not degraded beyond Ctrl (or patterns like earliest), SCSL definedness is too strong.
MP-ITA1–4	FIRST introduces	Continuation	Remainder	Continuation	Baselines	If SR does not

(tail continuations)	remainder variable t → licenses “some/a few/others” continuations.	existential independent; predicts no stable SR>Ctrl continuation advantage.	depends on discourse; not lexically forced; predicts smaller/variable advantage.	naturalness reliably higher in SR than Ctrl across languages and items.	lack FIRST-triggered remainder update, missing stable SR advantages.	increase continuation naturalness for “some others” vs earliest across languages, ITA is unsupported.
Entailment coding (succ/tail)	Higher entailment-strength in SR: successor and tail commitment are lexical/definedness.	Commitments defeasible/derived ; predicts weaker entailment deltas.	Can represent entailments but does not force tail existential from FIRST; predicts weaker tail commitment.	Mean coding(SR) > coding(Ctrl) for successor and tail; SR shows reduced cancellability.	Baselines do not predict robust tail-commitment increases triggered by FIRST.	If entailment coding does not differ SR vs Ctrl, the proposal reduces to a discourse preference rather than lexical structure.

8. Empirical methods and diagnostic set (replication-ready)

8.1 Participants and materials

A minimal judgment study can use native-speaker judgments (e.g., English N≈10–20; plus one additional language N≈6–12) or a compact corpus continuation task. Materials consist of minimal pairs that contrast FIRST (trigger) with a non-ordinal control (earliest) while holding the continuation constant.

8.2 Minimal pair set tailored to ITA

The following twelve items are tailored to the ITA prediction that FIRST preferentially licenses remainder indefinites (“some others”) and supports low-cost refinement to specific successors. For each pair, condition SR = ordinal-first trigger; condition Ctrl = non-ordinal control (“earliest”).

MP-ITA1 SR: John was first. Some others followed. | Ctrl: John arrived earliest. Some others followed.

MP-ITA2 SR: John was first. A few others followed. | Ctrl: John arrived earliest. A few others followed.

MP-ITA3 SR: John was first. Some others did too (ellipsis). | Ctrl: John arrived earliest. Some others did too.

MP-ITA4 SR: John was first. Some others followed—specifically, the second was Mary. | Ctrl: John arrived earliest. Some others followed—specifically, the second was Mary.

MP-ITA5 SR: John was first—there was no second position. | Ctrl: John arrived earliest—there was no second position.

MP-ITA6 SR: John was first—there were no other positions. | Ctrl: John arrived earliest—there were no other positions.

MP-ITA7 SR: If John is first, some others follow. | Ctrl: If John arrives earliest, some others follow.

MP-ITA8 SR: John might be first. Some others followed. | Ctrl: John might be earliest. Some others followed.

MP-ITA9 SR: John was first. The second was Mary. | Ctrl: John arrived earliest. The second was Mary.

MP-ITA10 SR: John was first. The third was Alex. | Ctrl: John arrived earliest. The third was Alex.

MP-ITA11 SR: John was first. Some others followed; the third was Alex. | Ctrl: John arrived earliest. Some others followed; the third was Alex.

MP-ITA12 SR: John was first. Some others followed, but none were second. | Ctrl: John arrived earliest. Some others followed, but none were second.

8.3 Coding protocol

Collect three measures per item: (i) acceptability (1–7), (ii) retrieval/continuation naturalness (1–7; “would you continue the discourse this way?”), and (iii) entailment-strength coding (0–2) for two targets: successor commitment (SECOND-slot availability) and tail remainder availability (“some others” licensing). Entailment coding: 0 = not supported; 1 = weak/defeasible; 2 = strong/non-cancellable.

8.4 Crosslinguistic extension: Turkish (typologically distinct)

A typologically distinct test case strengthens the universality claim by reducing the risk that ITA is an English-specific artifact. Turkish is useful because ordinals are morphologically transparent (birinci ‘first’, ikinci ‘second’, üçüncü ‘third’), while non-specific indefinites such as bazı ‘some’ and birkaç ‘a few’ are readily available. SCSL+ + ITA predicts that birinci should activate the successor spine and license remainder continuations that are naturally realized with bazı/birkaç plus diğerleri ‘others’. (Keenan & Stavi 1986; Kamp & Reyle 1993).

Illustrative diagnostics (adapt lexemes per consultant):

- (a) SR: Ali birinciydi. Bazı diğerleri geldi. ‘Ali was first. Some others came.’
- (b) Ctrl: Ali en erken geldi. Bazı diğerleri geldi. ‘Ali arrived earliest. Some others came.’
- (c) Cancellation: # Ali birinciydi ama ikinci sıra yoktu. ‘Ali was first but there was no second position.’
- (d) Embedding: Ali birinci olabilir. Bazı diğerleri gelmiş olabilir. ‘Ali might be first. Some others might have come.’

The key measurement parallels English: continuation naturalness (SR>Ctrl), strong degradation for cancellation in SR, and stronger entailment coding for successor/tail under the ordinal trigger.

8.5 Reporting template and planned analysis

This section specifies the planned analysis corresponding to the ITA minimal pairs. The primary reported quantities are mean(SR) – mean(Ctrl) by MP family for (i) acceptability and (ii) continuation/retrieval naturalness, plus (iii) entailment-strength deltas for successor and tail commitment (0–2 coding). Confidence intervals may be computed by participant bootstrap, or via a simple mixed-effects model with random intercepts for Participant and Item when sample size permits.

Appendix B provides the fillable matrices (B1–B3) and Appendix C provides a compact reviewer-facing template for summarizing predicted effects and for documenting the data acquisition route (consultants or corpus-based annotations).

8.6 Data acquisition options (feasibility)

Two concrete acquisition routes make the protocol immediately runnable. (i) Consultant judgments: English N=8 and Turkish N=6; 12 items; ~10 minutes per participant; report z-score–

normalized ratings per participant in addition to raw 1–7 scales. (ii) Corpus/LLM-elicitation as a supplement: collect naturally occurring or prompted continuations for the minimal pairs and annotate continuation types (“some/a few/others” vs alternatives). Such corpus/LLM evidence must be reported as exploratory and kept separate from human judgment statistics, but it can help validate item design and continuation inventories.

9. Computational and neurolinguistic implications

Computationally, SCSL+ + ITA yields an incremental update rule: encountering FIRST instantiates an index spine with a successor chain and introduces a remainder variable compatible with non-specific indefinites. This supports discourse inference about “others” and enables explicit RED refinements to specific successors. Neurolinguistically, the account predicts that FIRST triggers successor-chain activation and remainder availability, producing measurable differences in continuation readiness and in retrieval of successor identities relative to non-ordinal controls.

10. Limitations

Expanded versions are required here. Two issues should be addressed in expanded versions: (i) “first and only” contexts, which require explicit separation of index availability from occupancy and possibly repair/coercion rules; (ii) register/rhetorical uses of “first... no second,” which the theory predicts to involve contradiction and pragmatic repair rather than ordinary cancellation.

10.1 “First and only” / singleton contexts: index availability vs occupancy

A predictable objection is that natural discourse permits “first and only,” apparently contradicting successor closure. SCSL+ separates index availability (the successor spine in I) from occupancy (the partial mapping $f: I \rightarrow D$). Under this separation, $\text{succ}(\min(I))$ can be defined even if $f(\text{succ}(\min(I)))$ is undefined or contextually empty. Singleton readings can therefore be treated as cases where the successor slot exists but is unoccupied (or where the relevant domain restriction collapses occupancy while leaving the index spine intact). (Beaver 2001; Heim 1982). Where speakers explicitly insist on singletonhood (“first and only”), the analysis predicts either (i) a coercion/repair strategy that suppresses tail existential commitment while retaining successor availability, or (ii) a metalinguistic reading in which the ordinal is used

to mark onset without committing to filled successor slots. The important point is that the successor structure remains a lexical requirement on interpretability; what varies is how occupancy is accommodated.

10.2 Rhetorical uses (“first... no second”): contradiction and pragmatic repair

Another predictable objection is that speakers sometimes produce “first... but there was no second” rhetorically. SCSL+ + ITA treats these as contradiction/definedness clashes, not as successful implicature cancellation. Such utterances are therefore analyzed as pragmatic repair: the speaker exploits a clash with lexical definedness to achieve rhetorical emphasis, sarcasm, or metalinguistic correction. (Beaver 2001). This treatment is a feature rather than a bug: the theory’s goal is precisely not to preserve standard implicature diagnostics. If successor structure is lexical, then “cancellation” data should pattern like presupposition failure plus repair rather than like ordinary implicature withdrawal.

11. Conclusion

SCSL+ treats FIRST as a successor-saturating trigger: interpreting FIRST guarantees a successor at the initial index and makes the successor chain available as a structured tail. Indefinite Tail Activation adds a lexical interface rule: the tail is introduced as a non-specific remainder that is canonically lexicalized as “some/a few/others.” RED refines along successor; DED collapses refined successor identities back into remainder indefinites. The theory intentionally changes standard pragmatic diagnostics by moving successor and remainder effects from pragmatics into lexical definedness and interface update.

References

- Barwise, Jon & Robin Cooper. 1981. Generalized quantifiers and natural language. *Linguistics and Philosophy* 4: 159–219.
- Heim, Irene. 1982. The semantics of definite and indefinite noun phrases. PhD dissertation, University of Massachusetts Amherst.
- Ionin, Tania & Ora Matushansky. 2006. The composition of complex cardinals. *Journal of Semantics* 23(4): 315–360.

Kamp, Hans & Uwe Reyle. 1993. From Discourse to Logic. Dordrecht: Kluwer.

Keenan, Edward L. & Jonathan Stavi. 1986. A semantic characterization of natural language determiners. *Linguistics and Philosophy* 9: 253–326.

Partee, Barbara. 1987. Noun phrase interpretation and type-shifting principles. In *Studies in Discourse Representation Theory and the Theory of Generalized Quantifiers*, 115–143. Foris.

Yee, T. 2010. Building DRT lexical entries for superlatives and ordinal numbers. *Semantics Archive*.

Panner, Maks. 2026a. Successor-Closed Ordinals: A RED/DED Operator Account of Universal $FIRST \Rightarrow SECOND$ and Its Diagnostic Consequences. Manuscript.

Panner, Maks. 2026b. Cardinal–Ordinal Correlations as Bidirectional Discretization: A RED-Based Operator Account. Manuscript.

Beaver, David I. 2001. *Presupposition and Assertion in Dynamic Semantics*. Stanford, CA: CSLI Publications.

Kratzer, Angelika. 1991. Modality. In A. von Stechow & D. Wunderlich (eds.), *Semantics: An International Handbook of Contemporary Research*, 639–650. Berlin: de Gruyter.

Appendix A. Compact Elicitation Grid (ITA-focused)

Use the following grid across languages. Replace lexemes as needed while preserving the constructional contrasts.

Construction	Prompt	Target diagnosis
C1	“X is first.”	successor-chain + tail remainder introduced
C2	“X is first, but no second position exists.”	successor non-cancellability
C3	“X is first. Some others followed.”	ITA remainder licensing
C4	“X is first. The second was Y.”	RED refinement to successor identity
C5	“X is first. Some others followed; the second was Y.”	tail → successor refinement
C6	“Not first / Might be first / Is X first?”	projection under embedding

Appendix B. Results Matrices (fillable template; reporting format only)

No empirical claims are made from the values in Appendix B. The matrices are provided as a fillable measurement protocol and to illustrate the reporting format (participants $\times$ items) for the ITA minimal pairs. Replace the empty cells with ratings/codings from consultant judgments or another documented data source before interpreting any deltas or effect sizes.

Table B1. Acceptability matrix (1–7) for ITA minimal pairs (participants $\times$ items).

Scale: 1 = completely unacceptable; 7 = fully acceptable.

[illegible]

Table B2. Continuation / retrieval naturalness matrix (1–7) (participants $\times$ selected items).

Scale: 1 = very unnatural continuation; 7 = very natural continuation.

[illegible]

P3										
P4										
P5										
P6										
P7										
P8										
P9										
P10										
Mean										

Table B3. Entailment-strength coding matrix (0–2) for successor and tail commitments.

Scale: 0 = not supported; 1 = weak/defeasible; 2 = strong/non-cancellable.

Participant	Succ entail SR	Succ entail Ctrl	Tail entail SR	Tail entail Ctrl
P1				
P2				
P3				
P4				
P5				
P6				
P7				
P8				
P9				
P10				
Mean				

Appendix C. Aggregated reporting template (reviewer-facing)

Report mean(SR) – mean(Ctrl) for ITA items (e.g., MP-ITA1–MP-ITA4) and for cancellation items (MP-ITA5–MP-ITA6). Recommended summary: Δ acceptability, Δ continuation

naturalness, and Δ entailment strength for successor and tail. If N permits, report confidence intervals or a simple mixed-effects model with fixed effect Condition (SR vs Ctrl) and random intercepts for Participant and Item.